Article

Uncertainty-Aware Closed-Loop Model Predictive Control with EKF-SLAM for Safe Navigation of Nonholonomic Mobile Robots

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Abstract

Simultaneous Localization and Mapping (SLAM) supplies the state estimates a model predictive controller (MPC) needs, yet the two are usually combined under certainty equivalence, discarding SLAM's quantified uncertainty. We present a closed-loop framework in which an Extended Kalman Filter SLAM (EKF-SLAM) estimator feeds a constrained nonlinear MPC for a unicycle navigating an *unknown* environment: landmarks and obstacles enter the estimator once detected within a finite sensing range, and the posterior covariances of pose and sensed obstacles size the keep-out constraints online through a chance-constraint-inspired safety margin. In Monte-Carlo patrol simulation, SLAM reduces terminal error by 48% versus odometry while mapping to centimetre accuracy. Against a sensed static obstacle, fixed margins collide in 36% (odometry) and 14% (SLAM) of trials, whereas the covariance-aware margin recorded none (0 of 50) for under 1% additional path, and 0.50% (3 of 600) across thirty randomized geometries with its scaling factor frozen. Matched-margin ablations and a constant calibrated on disjoint training layouts match the adaptive margin (1 vs. 3 of 600): the demonstrated value is automatic online margin generation, not superior adaptivity. The mechanism extends to an intermittently visible moving obstacle (0 of 50 once stale tracks are dropped).

Keywords: mobile robotics; EKF-SLAM; model predictive control; obstacle avoidance; dynamic obstacle avoidance; covariance-scaled safety margins; uncertainty-aware control; nonholonomic systems

1. Introduction

Over the past two decades, interest in autonomous mobile robots has grown, driven by their potential across logistics, assistive care, and service operations. To act effectively in such settings, a robot must simultaneously determine its own location, build or exploit a map of its surroundings, follow a desired motion, and avoid obstacles while contending with sensor noise and actuator limits. Two technologies dominate the perception and control halves of this problem. Simultaneous Localization and Mapping (SLAM), and in particular Extended Kalman Filter SLAM (EKF-SLAM), provides a recursive, probabilistic estimate of the robot pose and landmark map together with its covariance [1,2]. Model Predictive Control (MPC), in turn, predicts the outcomes of candidate actions, optimizing the control inputs over a receding horizon while respecting constraints [3].

In practice, however, these two components are designed in isolation and stitched together under a *certainty-equivalence* assumption: the controller treats the SLAM estimate as though it were the true state, and the localization *uncertainty*, which the filter already

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computes explicitly, is discarded. Such an assumption is benign while the estimates remain accurate, yet it becomes the wrong one in the vicinity of obstacles, where a confident controller acting on an uncertain estimate may steer the *true* robot into a collision while its estimate indicates safety.

This paper sets out to close that gap in the setting SLAM was conceived for: an environment that is *unknown before it is sensed*. The map is not given: each landmark enters the estimator only once the robot first detects it within a finite sensing range, and the obstacles are likewise not specified in advance. Their positions (and, for a moving obstacle, velocities) are estimated online from the robot's own detections. We integrate this exploratory EKF-SLAM with a constrained nonlinear MPC for a unicycle robot and, rather than halting at certainty-equivalence, we feed the *covariances* of the pose and of every sensed obstacle back into the controller, so that the keep-out constraints widen when the estimates of the robot pose or of the sensed obstacles are uncertain, and tighten as that uncertainty decreases. The contributions of this study are as follows:

1. An end-to-end EKF-SLAM $\rightarrow$ NMPC pipeline for the unicycle model in an unknown environment: landmarks initialized from first detection under a finite sensing range, obstacles sensed and estimated online, and a constrained controller (actuator limits and obstacle avoidance) solved by a multiple-shooting transcription.
2. A *covariance-aware* obstacle-avoidance formulation: a chance-constraint-inspired safety margin that inflates each keep-out radius by a term proportional to the square root of the dominant eigenvalue of the combined pose and obstacle-estimate covariance.
3. A controlled Monte-Carlo study on a multi-leg patrol against an ideal-state oracle and a dead-reckoning baseline, quantifying tracking accuracy, map accuracy, collision rate, and the safety–efficiency trade-off, together with a randomized-geometry transfer study in which the frozen scaling factor reduces the collision rate roughly fifty-fold across thirty unseen, oracle-feasible layouts, under disjoint screening and evaluation seeds and continuous collision scoring.
4. An extension to *moving*, intermittently visible obstacles via a constant-velocity EKF tracker that coasts when the obstacle leaves the sensing range, including the track management this necessitates: without dropping stale tracks, the unboundedly growing prediction covariance trades mission completion for safety.
5. A matched-margin ablation, corrected after an instrumentation defect and re-run under a frozen estimand with disjoint calibration and evaluation seeds. With the arms matched to the declared active-step estimand (realised means agreeing to 0.3%), the experiment is *inconclusive* on whether temporal adaptation improves safety beyond a calibrated constant; what the covariance-aware shaping demonstrably provides is state-dependent margins without selecting a fixed constant directly.

2. Related Work

MPC for nonholonomic robots. Receding-horizon control of wheeled mobile robots is a mature research area. Howard et al. [4] demonstrated model-predictive control for tracking intricate paths under demanding mobility constraints, while Wang [5] lays out the systematic design of linear time-varying MPC about a reference trajectory. For the unicycle model in particular, Mehrez et al. [6] showed that a stabilizing nonlinear MPC (NMPC) can be solved online with open-source real-time optimization; it is this NMPC formulation that we adopt and extend with obstacle-avoidance constraints.

Output-feedback and estimation-in-the-loop. When the state cannot be measured directly, MPC is paired with an observer or filter and joined to it under certainty equivalence [3]. Robust and tube MPC instead address the estimation and disturbance error directly, tightening the constraints over a disturbance-invariant tube to guarantee their satisfaction, though at the price of constructing invariant sets [7]. Planning-under-uncertainty methods such as LQG-MP [8] go further, characterizing in advance the a priori state distribution that a chosen controller and sensor model induce, and selecting paths of low collision probability before execution ever begins. Our own approach is lighter than tube MPC and, unlike the offline analysis of LQG-MP, adapts *online* by reusing the covariance that the estimator already maintains and applying it only to the obstacle constraints most critical to safety.

SLAM and active/perception-aware planning. EKF-SLAM is a classical instrument of probabilistic robotics that maintains the joint pose–map covariance [1,2], and the broader arc of SLAM research is surveyed by Cadena et al. [9]. Active SLAM and perception-aware control—the perception-aware MPC of Falanga et al. [11] among them—reshape the robot's motion or its control objective in the service of better perception and estimation. Our approach differs in that we do not alter the estimation objective or seek information gain, but simply take the covariance the filter already yields and use it to make the *controller* safer, leaving the estimator untouched and confining the contribution to the control–constraint interface.

Chance-constrained MPC. Probabilistic (chance) constraints translate a bound on collision probability into a deterministic tightening of the constraint, by a multiple of the state-uncertainty standard deviation; Mesbah [12] surveys the stochastic-MPC landscape, and Blackmore et al. [13] approximate chance-constrained predictive control through particle and sampling methods. Nearest to our own obstacle treatment, Zhu and Alonso-Mora [14] formulate a real-time chance-constrained NMPC for collision avoidance in which the uncertainty arises from the predicted motion of dynamic obstacles. The same premise organises more recent work: Nair et al. [15] reformulate collision-avoidance constraints under uncertain obstacle predictions through Lagrange duality, obtaining deterministic tightenings for polytopic, Gaussian, and moment-based uncertainty descriptions, while Jian et al. [16] extend control barrier functions to moving obstacles and embed the resulting dynamic CBF in an MPC as a safety filter. Both treat uncertainty that originates in the *obstacle*, and both assume the robot's own pose is known; neither closes the loop on the estimator's pose covariance, and a CBF safety filter is a complementary mechanism to the margin studied here rather than a competing one. We adopt the same constraint-tightening device but drive it from a different and underused source—the robot's own online EKF-SLAM covariance, combined with the covariance of each obstacle estimate it maintains—so that the safety margin follows the live quality of both localization and perception rather than worst-case bounds.

Joint robot-and-environment uncertainty. Several strands of prior work combine subsets of the elements considered here. Chance-constrained MPC formulations bound collision risk under Gaussian obstacle uncertainty in real time [18], in some cases jointly with robot-state uncertainty through the overlap of the two distributions [19], and recent sampling-based variants propagate both state and obstacle uncertainty through the prediction horizon with probabilistic chance constraints [20]; in all three, however, obstacle estimates are supplied by an external tracker or an a priori map, and the robot pose is not estimated jointly with the environment. Belief-space approaches account for robot, landmark, and obstacle uncertainty simultaneously and derive collision probabilities under Gaussian assumptions [21], while online-mapping frameworks incrementally build an uncertainty-aware environment representation and replan probabilistically safe trajectories in unknown environments [22]; both, though, rely on planning layers decoupled from the controller rather than on closed-loop NMPC, and neither feeds the correlated covariance of a SLAM filter running in the loop into the avoidance constraints. Conversely, the active-SLAM literature uses receding-horizon control to reduce estimation uncertainty, but with exploration or information objectives rather than safety constraints [23]. The contribution of this paper is therefore not covariance-based constraint inflation in isolation, which the works above employ in various forms, but the integration and its audit: keep-out constraints inflated online with covariances from an exploratory EKF-SLAM filter in which both landmarks and obstacles are discovered during the mission, obstacle tracks maintained under intermittent visibility, and the closed loop evaluated for transfer across thirty randomized layouts with the inflation gain frozen and screening seeds disjoint from evaluation seeds, including robustness tests under deliberate filter mis-calibration.

3. System Models

3.1. Unicycle Kinematic Model

The robot is modeled as a unicycle with pose $\mathbf{x}_{r,t} = [x_t, y_t, \theta_t]^\top$ and control $\mathbf{u}_t = [v_t, \omega_t]^\top$, where v_t is the linear velocity and ω_t the angular velocity. The continuous-time kinematics are

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta, \qquad \dot{\theta} = \omega. \tag{1}$$

For control we use the Euler discretization at sampling period Δt,

$$\mathbf{x}_{r,t+1} = \mathbf{x}_{r,t} + \Delta t \begin{bmatrix} v_t\cos\theta_t \\ v_t\sin\theta_t \\ \omega_t \end{bmatrix}, \tag{2}$$

while for the EKF-SLAM prediction we use the exact integral of the constant-twist motion over Δt, which avoids the linear-arc error of (2) for $\omega \neq 0$.

3.2. Range–Bearing Measurement Model

The environment contains M point landmarks with coordinates $(l_{x,i}, l_{y,i})$. The robot observes each landmark through a range–bearing sensor,

$$\mathbf{z}_t^i = \begin{bmatrix} d_t^i \\ \phi_t^i \end{bmatrix} = \begin{bmatrix} \sqrt{(l_{x,i}-x_t)^2 + (l_{y,i}-y_t)^2} \\ \operatorname{atan2}(l_{y,i}-y_t,\ l_{x,i}-x_t) - \theta_t \end{bmatrix} + \mathbf{v}_t^i, \qquad \mathbf{v}_t^i \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \tag{3}$$

with $\mathbf{R} = \operatorname{diag}(\sigma_d^2, \sigma_\phi^2)$. Bearings are wrapped to $(-\pi, \pi]$. The sensor has a finite range r_{sense}: a landmark (or obstacle) is observed only while it lies within r_{sense} of the robot, so the environment is *unknown before it is detected*. Landmarks are assumed distinguishable (known data association), as with uniquely identifiable features or beacons.

4. Methodology

The augmented EKF-SLAM state stacks the robot pose and the M landmarks, $\mathbf{x}_t = [\mathbf{x}_{r,t}^\top, l_{x,1}, l_{y,1}, \ldots, l_{x,M}, l_{y,M}]^\top \in \mathbb{R}^{3+2M}$, estimated by mean $\hat{\mathbf{x}}_t$ and covariance $\mathbf{P}_t$.

4.1. EKF-SLAM

We use the standard EKF-SLAM formulation [1]. At each step the filter performs a prediction using the commanded control $\mathbf{u}_t$ (the filter does not observe the actuation noise, so process uncertainty is genuine) followed by a measurement update. The prediction propagates the pose with the exact velocity-motion model and inflates the covariance through the control-to-state Jacobian V_t and input-noise covariance $\mathbf{Q}_u = \operatorname{diag}(\sigma_v^2, \sigma_\omega^2)$:

$$\hat{\mathbf{x}}_t^- = \hat{\mathbf{x}}_{t-1} + F_x^\top g(\hat{\mathbf{x}}_{t-1}, \mathbf{u}_t), \qquad \mathbf{P}_t^- = G_t\,\mathbf{P}_{t-1}\,G_t^\top + F_x^\top\,(V_t \mathbf{Q}_u V_t^\top)\,F_x, \tag{4}$$

where F_x maps the pose block into the full state and G_t is the motion Jacobian. The update is initialized with the prior, $\hat{\mathbf{x}}_t \leftarrow \hat{\mathbf{x}}_t^-$ and $\mathbf{P}_t \leftarrow \mathbf{P}_t^-$, and landmarks are processed *sequentially*: for each observed landmark i the predicted measurement $\hat{\mathbf{z}}_t^i$ and Jacobian H_t^i are computed from the current *estimated* landmark (no ground-truth information is used), the innovation bearing is wrapped, and the standard Kalman correction is applied before processing the next landmark:

$$K_t^i = \mathbf{P}_t H_t^{i\top} \left(H_t^i \mathbf{P}_t H_t^{i\top} + \mathbf{R}\right)^{-1}, \quad \hat{\mathbf{x}}_t \leftarrow \hat{\mathbf{x}}_t + K_t^i\big(\mathbf{z}_t^i - \hat{\mathbf{z}}_t^i\big), \quad \mathbf{P}_t \leftarrow (I - K_t^i H_t^i)\,\mathbf{P}_t. \tag{5}$$

The map is *unknown a priori*. Each landmark is initialized from its first in-range observation through the inverse measurement model, $\hat{l}_{x,i} = \hat{x} + d\cos(\hat{\theta} + \phi)$, $\hat{l}_{y,i} = \hat{y} + d\sin(\hat{\theta} + \phi)$, under an uninformative prior, and is refined by every subsequent re-observation; landmarks outside the sensing range contribute no update, so the pose covariance grows through landmark-poor stretches and contracts on re-detection. We adopt the standard *start-frame* convention of exploratory SLAM: the robot's initial pose defines the world frame and carries zero initial uncertainty. This fixes the gauge freedom of unknown-map SLAM (a rigid transformation of robot and map together leaves every range–bearing measurement unchanged, so a pose in an externally defined frame is unobservable), and it makes navigation goals well posed as coordinates *relative to the deployment point*, which is how a robot without external infrastructure operates in practice. In this setting the benefit of SLAM over dead reckoning is *drift suppression*. Odometric error grows without bound along the mission, whereas re-observing self-mapped landmarks bounds it.

4.2. Constrained Nonlinear MPC

At each sampling instant the controller solves, over horizon N,

$$\begin{aligned}
\min_{\mathbf{X},\mathbf{U}} \; & \sum_{k=0}^{N-1} \Big[(\mathbf{x}_{r,k} - \mathbf{x}^\star)^\top \mathbf{W}_x (\mathbf{x}_{r,k} - \mathbf{x}^\star) + \mathbf{u}_k^\top \mathbf{W}_u\, \mathbf{u}_k\Big] \\
\text{s.t. } & \mathbf{x}_{r,0} = \hat{\mathbf{x}}_{r,t}, \qquad \mathbf{x}_{r,k+1} = \mathbf{x}_{r,k} + \Delta t\, f(\mathbf{x}_{r,k}, \mathbf{u}_k), \\
& v_{\min} \le v_k \le v_{\max}, \quad \omega_{\min} \le \omega_k \le \omega_{\max}, \\
& \|\mathbf{p}_k - \mathbf{o}_j\| \ge r_{\text{rob}} + r_j + \delta, \quad j = 1, \ldots, N_o,\; k = 0, \ldots, N,
\end{aligned} \tag{6}$$

where $\mathbf{x}^\star$ is the active reference posture, $\mathbf{x}_{r,k}$ the predicted pose over the horizon (index k), $\mathbf{p}_k = [x_k, y_k]^\top$, $\mathbf{o}_j$ and r_j the j-th obstacle center and radius, r_{rob} the robot radius, and $\delta \ge 0$ a safety margin defined below. The initial state is set to the pose block $\hat{\mathbf{x}}_{r,t}$ of the current EKF-SLAM estimate. Missions are specified as an ordered list of waypoints in the start frame; a simple mission executive supplies the active waypoint as $\mathbf{x}^\star$, advancing when the estimate arrives within a pass-through tolerance or a per-waypoint timeout expires (the timeout prevents an erroneous estimate hovering just outside the tolerance from wedging the mission; the estimation error still registers in the terminal metric). The problem is transcribed by *multiple shooting* (states and controls are decision variables, dynamics enter as equality constraints), which improves convergence of the nonconvex obstacle constraints, and solved with an interior-point method [6]. Only the first control is applied; the horizon is then shifted and warm-started.

4.3. Closed-Loop Coupling

The closed loop runs at period Δt: EKF-SLAM produces $(\hat{\mathbf{x}}_t, \mathbf{P}_t)$; the NMPC (6) consumes the pose estimate $\hat{\mathbf{x}}_{r,t}$ (and, below, $\mathbf{P}_t$) and returns $\mathbf{u}_t$; the actuated control $\mathbf{u}_t + \mathbf{w}_t$ drives the true plant; the new range–bearing measurement updates the filter. Under certainty equivalence the controller would use $\hat{\mathbf{x}}_{r,t}$ only; our contribution additionally uses $\mathbf{P}_t$.

4.4. Sensed Obstacles and the Covariance-Aware Safety Margin

Consistent with the unknown-map setting, obstacles are not declared to the controller; they must be *discovered* at runtime. A static obstacle is estimated by a two-state EKF driven by the robot's range–bearing detections: the estimate $\hat{\mathbf{o}}$ is initialized from the first in-range detection through the inverse measurement model, with an initial covariance Σ_o that inherits the robot's pose uncertainty through the corresponding Jacobians, and each subsequent update folds the current pose covariance into an effective measurement noise $\mathbf{R}_{\text{eff}} = \mathbf{R} + H_p \mathbf{P}_{\text{pose}} H_p^\top$, keeping Σ_o consistent without a joint filter. Until the obstacle is first detected, the keep-out constraint in (6) is inactive: the controller cannot avoid an undetected obstacle, and the sensing range must therefore provide sufficient reaction distance.

A fixed margin $\delta = \delta_0$ cannot account for time-varying estimation quality. We therefore size the margin from the live covariances. Let $\mathbf{P}_{xy}$ be the 2×2 position block of $\mathbf{P}_t$, Σ_o^{pos} the position block of the obstacle-estimate covariance, and $\lambda_{\max}(\cdot)$ the largest eigenvalue. We set

$$\delta = \delta_0 + \gamma \sqrt{\lambda_{\max}\big(\mathbf{P}_{xy} + \Sigma_o^{\text{pos}}\big)}, \tag{7}$$

where δ_0 is a fixed buffer (chosen so that an ideal-state controller recorded no collision at this discretization) and $\gamma \geq 0$ scales the tightening in the manner of chance-constrained MPC, in which constraints are tightened by a multiple of the state standard deviation to bound the violation probability [12,13]. (7) is chance-constraint-*inspired* rather than a calibrated probability bound: linearization error, non-Gaussianity, and correlations across horizon nodes preclude an exact guarantee, so γ is selected empirically from the safety–efficiency sweep reported in the Results. The square-root term is the relative-position uncertainty along its worst-case direction, so the keep-out radius grows when, and in the direction in which, the relative-position uncertainty is largest. Equation (7) is available only because the filters provide their covariances online; a dead-reckoning controller has no comparable uncertainty estimate to act on.

4.5. Extension to Dynamic Obstacles and Track Management

The framework extends to a *moving* obstacle without changing the estimator or controller structure. The obstacle is tracked by a constant-velocity (CV) EKF from the robot's range–bearing detections, producing a position–velocity estimate $\hat{\mathbf{o}}_t$ and covariance $\Sigma_{o,t}$, in the spirit of chance-constrained dynamic avoidance [14]. As with the static case, the track is created on the first in-range detection with a covariance that inherits the pose uncertainty (the initial velocity is unknown and receives a large prior). Because the sensing range is finite, visibility is *intermittent*: while the obstacle is out of range the tracker *coasts*, predicting without correction, and its covariance grows. Over the horizon the obstacle is propagated by the CV model to $\hat{\mathbf{o}}_{t+k|t}$ with covariance $\Sigma_{o,k}$ that *grows* with the look-ahead k. The keep-out constraint in (6) becomes time-varying,

$$\|\mathbf{p}_k - \hat{\mathbf{o}}_{t+k|t}\| \geq r_{\text{rob}} + r_{\text{obs}} + \delta_k, \tag{8}$$

and the covariance-aware margin combines both uncertainty sources,

$$\delta_k = \delta_0 + \gamma \sqrt{\lambda_{\max}\big(\mathbf{P}_{xy} + \Sigma_{o,k}^{\text{pos}}\big)}, \tag{9}$$

where $\Sigma_{o,k}^{\text{pos}}$ is the position block of the predicted obstacle covariance. The keep-out bubble thus inflates both because the robot is unsure of its own pose and because the predicted obstacle position is uncertain, increasingly so at longer look-ahead. Two properties of (9) should be stated plainly. First, the obstacle covariance is propagated along the horizon but the robot's pose covariance enters at its *current* value $\mathbf{P}_{xy}$ at every node: the margin is a current-covariance heuristic, and it may understate the relative uncertainty at longer look-ahead during landmark-poor or aggressively evasive motion. Second, to preserve feasibility in tight encounters the obstacle constraint is softened. One non-negative slack $\sigma_k \geq 0$ per horizon node relaxes the keep-out inequality and the cost carries the quadratic penalty $\rho \sum_k \sigma_k^2$ with $\rho = 10^3$, three orders of magnitude above the tracking weights, so the slack stays inactive unless the hard constraint is unreachable; a regulariser of 10^{-4} inside the distance square root keeps the constraint gradient finite at coincident positions. Softening means the optimizer may in principle accept a predicted incursion rather than report infeasibility, so the keep-out is not a hard guarantee. We therefore report the *realised* minimum clearance, judged on true geometry, rather than assuming the constraint binds; in the covariance-aware runs it remained positive in every trial reported here.

Intermittent visibility requires one further component, track management. A track that has not been re-detected for a staleness window is dropped, and the constraint deactivates until re-detection. Without it, the coasting covariance grows without bound and (9) inflates a stale keep-out bubble that can block the workspace. In our experiments the un-managed tracker kept collisions near zero but left the mission unfinished in many trials. This is a structural property of uncertainty-aware margins under partial observation, since the margin is only as reasonable as the lifetime of the estimate on which it depends.

In summing the two covariances in (9) we treat the robot and obstacle uncertainties as independent. The two are in fact coupled—the obstacle is detected from the robot's own uncertain pose—and this coupling is expected to correlate the two errors *positively*; positively correlated errors partially cancel in the relative position, so summing the marginals over-approximates the relative-position uncertainty. This is a physical argument rather than a proof. The exact relative covariance is $\mathbf{\Sigma}_{\text{rel}} = \mathbf{\Sigma}_r + \mathbf{\Sigma}_o - \mathbf{\Sigma}_{ro} - \mathbf{\Sigma}_{ro}^\top$, and dropping the cross terms is conservative only where $\mathbf{\Sigma}_{ro} + \mathbf{\Sigma}_{ro}^\top$ is positive semidefinite along the direction that sets the margin. We do not verify this directionally, so the independence approximation should be read as an assumption of the present formulation rather than an established bound; a joint robot–obstacle filter would recover the cross-covariance at the price of added complexity.

5. Simulation Setup

Scenario. The robot executes a return-to-start *patrol*: from the start pose $(0, 0, 0)$, which defines the world frame, it visits the waypoints $(1.5, 1.5)$ and $(-1.5, 1.0)$ and returns to the origin, a mission of roughly 7 m. Three landmarks, unknown to the robot beforehand, lie at $(-0.5, 0.5)$, $(1, 1)$, and $(-1.5, 0)$; the sensing range is $r_{\text{sense}} = 1.2$ m, chosen so that sensing coverage varies along the mission: only the first landmark is in range at the start and the rest are discovered en route, while the number of landmarks in range varies between one and two along the route, and the pose uncertainty peaks near the waypoints where the robot turns; this nonstationarity is what the covariance-aware margin is designed to track. (In the executed trials the obstacle passage itself is the best-covered part of the mission—two of three landmarks in range in all 50 trials—with pose uncertainty at roughly half its mission peak; the margin follows the live covariance wherever the mission takes it, not a designated danger zone.) For the static-obstacle case a disk of radius 0.15 m sits at $(0, 1.25)$, on the second leg; it is encountered after drift has accumulated and is itself unknown until detected. The robot radius is 0.15 m. Actuator limits are $v_{\max} = 0.6$ m/s, $\omega_{\max} = \pi/4$ rad/s. The free-space controller uses $N = 10$, $\Delta t = 0.05$ s; the obstacle controller uses $N = 14$, $\Delta t = 0.2$ s for adequate lookahead. Cost weights are $\mathbf{W}_x = \mathrm{diag}(1, 5, 0.1)$, $\mathbf{W}_u = \mathrm{diag}(0.5, 0.05)$. The mission executive uses a 0.10 m pass-through tolerance and a per-waypoint timeout for intermediate waypoints and a 0.05 m tolerance at the final goal. The timeout is 15 s in the static studies and 30 s (with an extended 75 s mission budget) in the dynamic scenario: a safe controller must be allowed to wait out a crossing obstacle rather than being forced to abandon a temporarily blocked waypoint, while the timeout still bounds the pathological case of an erroneous estimate hovering just outside the arrival tolerance.

Noise. Range and bearing measurement noise have variance $\sigma_d^2 = \sigma_\phi^2 = 0.01$ (standard deviations 0.1 m and 0.1 rad); actuation noise on v and ω has variance 0.01 (standard deviation 0.1), a level at which dead-reckoning drift is material over the mission, the regime for which SLAM is intended. The filter is given the same noise parameters (a consistent filter). There is no initial pose offset: the start pose defines the frame and carries zero uncertainty, so all localization error is *accumulated*, not injected. The fixed buffer is $\delta_0 = 0.06$ m and the scaling factor is $\gamma = 2$ unless swept. In the dynamic scenario the obstacle starts at $(1.3, 0.15)$ with crossing velocity $(-0.16, 0.16)$ m/s and true acceleration noise of standard deviation 0.05, a moderately manoeuvring adversary whose speed remains below the robot's over the 75 s mission; the tracker assumes a larger acceleration variance (0.02), so its coasting predictions err on the conservative side. Tracks are dropped after 3 s without a detection: by then the coasting position uncertainty ($\approx$0.4 m at the assumed acceleration noise) exceeds the physical keep-out radius, and the prediction is no longer actionable.

Baselines. Feedback sources and obstacle strategies are compared on *identical* noise realizations per trial: *oracle* (the NMPC is given the true pose and, in obstacle cases, the true obstacle; an upper performance bound), *odom* (dead reckoning: EKF prediction only, no measurement correction), and *slam* (full EKF-SLAM). Sharing the noise across modes isolates the effect of the estimator or margin under test.

Metrics. We report the *terminal true-vs-reference error* (the distance between the *true* final pose and the final waypoint, that is, what the robot actually achieves, not what its estimate reports), the mean localization error, the final *map error* (mean distance between estimated and true landmark positions), the collision rate and minimum obstacle clearance judged on true geometry, and the path length. We run $N_{\mathrm{MC}} = 50$ Monte-Carlo trials and report means with 95% confidence intervals (Wilson intervals for collision proportions). Collision statistics for the primary studies pertain to the fixed scenario geometries described above, on which γ is also selected by the sweeps; Section 6.6 tests whether the frozen operating point transfers to layouts not seen during its selection.

Implementation. The framework is implemented in MATLAB. The constrained NMPC is formulated and solved with CasADi [17] (v3.5.5) using its bundled IPOPT interior-point solver, whereas the EKF-SLAM estimator, the static obstacle estimator, and the constant-velocity tracker are implemented directly in MATLAB.

6. Results

6.1. Tracking and Mapping on the Free-Space Patrol

We begin without obstacles. Table 1 and Figure 1 gather the patrol study. Dead reckoning drifts as the mission lengthens, ending 0.122 m from the final waypoint with a mean localization error of 0.083 m; closing the loop with EKF-SLAM pulls the terminal error down to 0.064 m (a 48% reduction, with non-overlapping confidence intervals) and halves the localization error to 0.041 m, while simultaneously building the landmark map *from scratch* to a final accuracy of 0.040 m. SLAM lands within 0.017 m of the oracle bound. The uncertainty trace (Figure 1, right) shows the expected uncertainty signature:

the pose covariance swells through the landmark-poor mid-tour and contracts on each re-detection, whereas the dead-reckoning covariance only grows. The return-to-start leg additionally re-observes the landmarks mapped at the outset, a loop closure in miniature, which pulls SLAM's terminal error back to the oracle bound while odometry, having no such mechanism, retains its accumulated drift.

Table 1. Free-space patrol Monte-Carlo results ($N_{\mathrm{MC}} = 50$, mean $\pm$ 95% CI). Map error is the final mean landmark-position error (SLAM builds the map online; the others maintain none).

Feedback	Terminal Error [m]	Localization Error [m]	Map Error [m]
oracle	0.0466 ± 0.0008	0.0000	—
odom	0.1224 ± 0.0179	0.0829 ± 0.0105	—
slam	$\mathbf{0.0641 \pm 0.0093}$	$\mathbf{0.0410 \pm 0.0059}$	**0.0403**

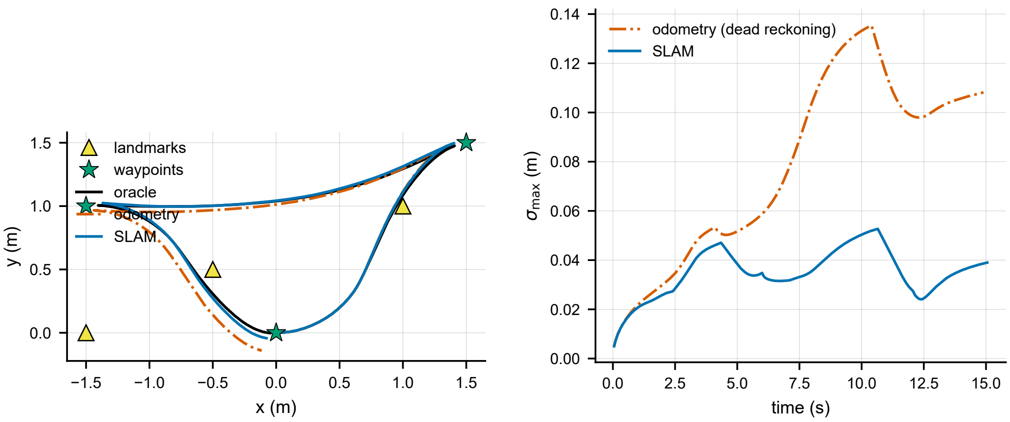

Figure 1. Free-space patrol. (**Left**) true trajectories for one trial; (**Right**) pose-uncertainty evolution $\sigma_{\max}$ over the tour: SLAM's uncertainty contracts on each landmark re-detection while dead reckoning's only grows.

6.2. Sensed Static Obstacle: Fixed vs. Covariance-Aware Margin

We now place the obstacle on the second leg, unknown to the robot until detected. With a *fixed* margin δ_0, the clairvoyant oracle never collides, but odometry collides in 36% of trials (worst penetration -0.171 m) and SLAM in 14% (Table 2). SLAM's better estimate helps—its collision rate is less than half of odometry's, and its obstacle estimate converges to 0.068 m—but a fixed margin cannot exploit the fact that SLAM can quantify when it is uncertain. (The passage does not, in this geometry, coincide with the pose covariance's peak—the stored trials put it near a local minimum—so the case for the covariance-aware margin here is not that the obstacle sits in the worst spot, but that the margin needs no knowledge of where the worst spot is.)

Activating the covariance-aware margin (7) with $\gamma = 2$ records no SLAM collision (0 of 50 trials, worst clearance $+0.073$ m) with no detected terminal-error difference (0.089 m vs. 0.091 m). Because all strategies share noise seeds, the collision contrasts admit exact paired tests: every discordant pair favours the covariance-aware margin (0:18 against odometry, McNemar $p = 7.6 \times 10^{-6}$; 0:7 against the SLAM fixed margin, $p = 0.016$), with paired risk reductions of 36 pp [22, 50] and 14 pp [6, 24]. The added safety costs almost nothing: the mean path grows by less than 1% (6.99 m vs. 6.93 m). The inflation is in fact nonzero on most steps (93.6%), but it is small in absolute terms (0.07–0.15 m in a 4×4 m workspace) and the obstacle constraint binds the trajectory only when an obstacle is nearby, so inflation elsewhere costs no path. Because the noise seeds are shared, the oracle and odometry columns are identical across margin settings, confirming that the only change is the margin.

Figure 2 shows the trajectories and the margin dynamics: the applied inflation peaks at first detection, when the freshly initialized obstacle estimate is most uncertain, and again near the waypoints where pose uncertainty is largest; at the obstacle passage itself it sits near a local minimum.

Table 2. Sensed-obstacle Monte-Carlo results ($N_{\mathrm{MC}} = 50$; Wilson 95% CIs). All non-oracle strategies estimate the obstacle online from detections.

Strategy	Collision Rate (95% CI)	Worst Clearance [m]	Terminal Error [m]	Path [m]
oracle (clairvoyant)	0% [0.0, 7.1]	+0.023	0.034	6.92
odom + fixed δ_0	36% [24.1, 49.9]	−0.171	0.201	6.85
slam + fixed δ_0	14% [7.0, 26.2]	−0.058	0.091	6.93
slam + covariance-aware	**0% [0.0, 7.1]**	**+0.073**	**0.089**	**6.99**

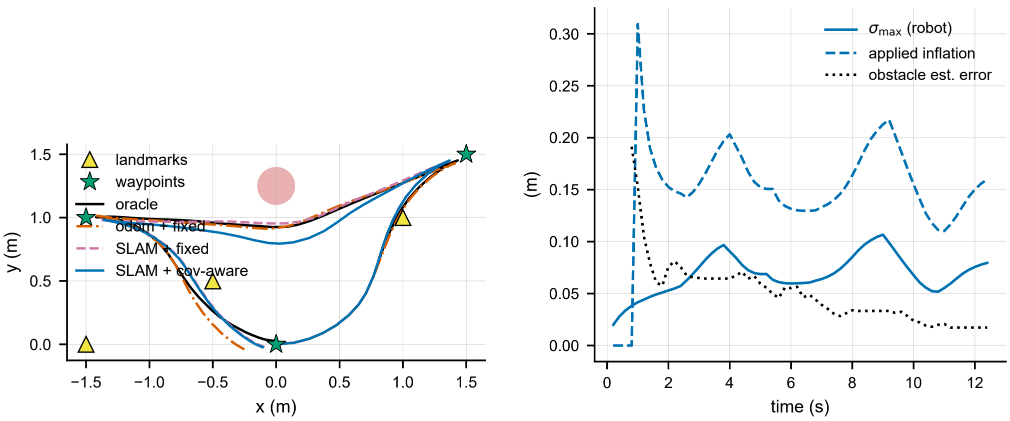

Figure 2. Sensed static obstacle. (**Left**) true trajectories for one trial; (**Right**) covariance-aware margin dynamics over the same trial: the applied inflation initializes at first detection and follows the pose and obstacle-estimate uncertainty; in this geometry it peaks at first detection and near the waypoints rather than at the obstacle passage.

6.3. *Safety–Efficiency Trade-off*

Figure 3 and Table 3 sweep the scaling factor γ. The collision rate falls monotonically from 14% at $\gamma = 0$ to 2% at $\gamma = 0.5$ and is 0% for every $\gamma \geq 1$, while terminal error remained approximately 0.09 m with no evident trend. The safety costs little across this sweep: the mean path grows by only 1.5% across the entire range (6.93 m at $\gamma = 0$ to 7.04 m at $\gamma = 3$), and the worst-case clearance climbs steadily (−0.058 to +0.141 m). Because the margin keys off the live covariance, the nominal inflation is state-dependent rather than constant; its absolute size remains small everywhere in this workspace, which is why the path cost stays low across the sweep. The sweep justifies the operating point $\gamma = 2$: it lies well inside the range of γ over which no collision was observed, with headroom against sampling variability.

Table 3. Scaling-factor (γ) sweep ($N_{MC} = 50$, sensed static obstacle). $\gamma = 0$ is the fixed margin δ_0.

γ	**Collision Rate**	**Path Length [m]**	**Terminal Error [m]**	**Worst Clearance [m]**
0.0	14%	6.93	0.091	−0.058
0.5	2%	6.95	0.092	−0.026
1.0	0%	6.96	0.088	+0.007
1.5	0%	6.98	0.090	+0.042
2.0	0%	6.99	0.089	+0.073
2.5	0%	7.01	0.085	+0.106
3.0	0%	7.04	0.084	+0.141

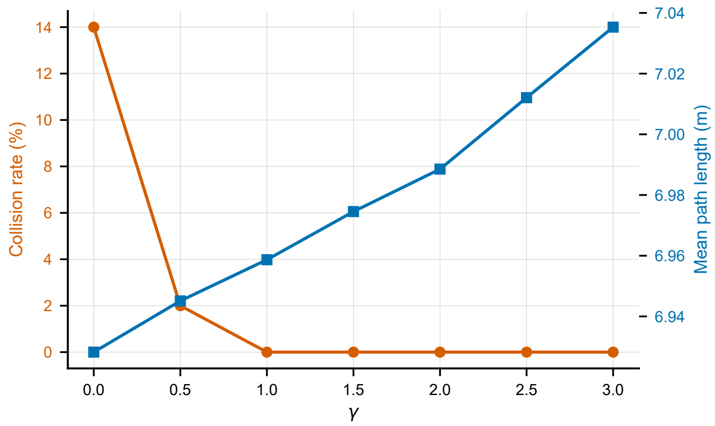

Figure 3. Safety–efficiency trade-off of the covariance-aware margin (sensed static obstacle). Collision rate (left axis) falls monotonically to zero as γ grows; mean path length (right axis) increases by less than 2% across the whole range.

6.4. Dynamic Obstacles under Intermittent Visibility

We now let the obstacle move, crossing the first patrol leg with random accelerations, so a constant-velocity prediction cannot be exact, and, because the sensing range is finite, the obstacle is repeatedly *lost and re-acquired* as robot and obstacle move apart and back. Four strategies are compared: *oracle* (true pose and clairvoyant true obstacle trajectory), *static* (freeze the obstacle at its current estimate), *cv_fixed* (CV-predicted track, fixed margin), and *cv_cov* (CV-predicted track, covariance-aware margin (9), $\gamma = 2$). Robot localization is EKF-SLAM in all non-oracle modes, so the comparison isolates the obstacle-handling strategy.

Intermittent visibility makes this scenario far harsher than a fully observed one: Table 4 shows that freezing the obstacle collides in 88% of trials and CV prediction with a fixed margin still collides in 44%—the prediction helps, but a fixed margin cannot absorb the error a coasting track accumulates. The covariance-aware margin recorded no collision (0 of 50 trials, worst clearance +0.210 m), the same observed count as the clairvoyant oracle—equal zero counts are not an equivalence result—at the cost of a substantially more cautious path (15.4 m vs. the oracle's 7.0 m): with only intermittent contact, the added caution appears as detours. The paired collision contrasts are 0:44 discordant against the frozen-estimate strategy (McNemar $p = 1.1 \times 10^{-13}$) and 0:22 against CV-fixed ($p = 4.8 \times 10^{-7}$), risk reductions 88 pp [78, 96] and 44 pp [30, 58]; all four collision contrasts in this paper survive Holm correction as one declared family.

The detours carry a second, subtler cost that the collision and terminal metrics do not expose, and Table 4 therefore reports *true* waypoint achievement: the number of waypoints the true robot passes within 0.15 m of (one robot radius), and the mean true miss distance.

The covariance-aware controller strictly visits only 2.0 of 3 waypoints (mean miss 0.11 m), fewer than the colliding fixed strategies (2.4–2.5). The cause is not that the executive abandons blocked waypoints: in a representative missed trial the *estimate* arrived 0.07 m from the waypoint —within tolerance—while the true robot passed 0.48 m away, the localization error having grown to 0.41 m at that moment (against 0.04 m in the free-space patrol). The safety detour itself degrades localization: evading the obstacle takes the robot away from the landmarks through aggressive manoeuvres where drift compounds, and the waypoints are then visited in the drifted belief frame, with the estimate re-anchoring only on the homeward leg (hence the healthy terminal error). The observed safety outcome is unaffected—collisions are judged on true geometry, and the realised minimum clearance remained positive in every trial despite the drift the detours introduce. The fixed strategies post better waypoint numbers only because they do not detour, and they collide in 44–88% of trials doing so.

Read at the *mission* level, the comparison is therefore not close, because a collided patrol is a failed patrol. The fixed strategies buy their extra waypoint precision on missions that end in collision 44–88% of the time, whereas the covariance-aware controller finished every patrol without contact, at a mean true miss distance of 0.11 m, though it passed within the 0.15 m threshold of only 2.00 of the 3 waypoints on average—the detours buy safety at a cost in true-frame waypoint precision, and neither strategy dominates on both.

Track management proved to be *necessary* rather than optional. Without the 3 s staleness drop, the covariance-aware controller recorded at most a single collision but its coasting track's covariance grew without bound, inflating a stale, oversized exclusion region that blocked the workspace; in an early configuration of this scenario (a more strongly manoeuvring obstacle, acceleration noise standard deviation 0.15, before the staleness drop was introduced) the mean terminal error ballooned beyond 2.5 m because many patrols simply could not be completed. Dropping stale tracks restores mission completion at unchanged safety, which shows that an uncertainty-aware margin is only as useful as the lifetime of the estimate that supplies it.

Table 4. Dynamic-obstacle results ($N_{\mathrm{MC}} = 50$, crossing obstacle, intermittent visibility, track management active). Waypoints visited counts true passes within 0.15 m; miss distance is the mean true closest approach over the three waypoints.

Strategy	Collision (95% CI)	Worst Clear. [m]	Path [m]	Wps Visited	Miss [m]
oracle (clairvoyant)	0% [0.0, 7.1]	+0.002	6.95	3.00/3	0.05
static (ignore motion)	88% [76.2, 94.4]	−0.268	7.01	2.42/3	0.09
cv_fixed	44% [31.2, 57.7]	−0.179	7.29	2.48/3	0.08
cv_cov (cov-aware)	**0% [0.0, 7.1]**	+**0.210**	15.44	2.00/3	0.11

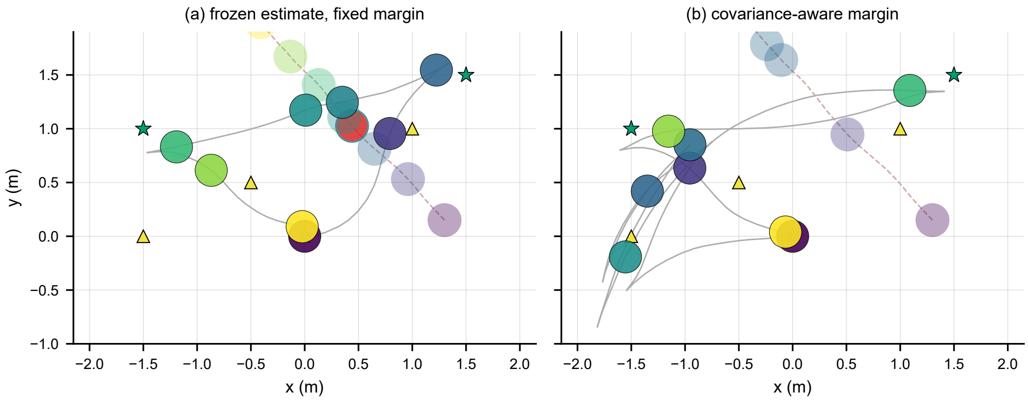

Figure 4. Dynamic obstacle under intermittent visibility, same trial (identical obstacle motion). Robot (solid) and obstacle (translucent) are drawn as disks ($r = 0.15$ m) at evenly spaced instants; simultaneous snapshots share a colour, and the obstacle disk turns red on overlap. (**a**) Frozen estimate with fixed margin: the disks overlap mid-crossing (collision). (**b**) Covariance-aware margin: the robot detours south of the coasting prediction and the simultaneous pairs never meet.

Figure 5 sweeps γ in the dynamic scenario. The collision rate collapses from 44% at $\gamma = 0$ to 2% at $\gamma = 0.5$ and is 0% for every $\gamma \geq 1$. Unlike the static case, the caution here is expensive: the path grows from 7.3 m to 15.7 m across the sweep, with a clear jump between $\gamma = 0.5$ (9.0 m) and $\gamma = 1$ (14.4 m), because intermittent contact forces wide detours around coasting predictions. This suggests an operating-point observation for the simulated regimes studied here (finite Monte-Carlo counts cannot support a deployment safety recommendation): where collisions are unacceptable and path budget is ample, adopt the *more conservative* setting $\gamma = 2$, which sits well inside the region over which no collision was observed in these trials, the plateau in both scenarios; where efficiency matters and a small residual risk is tolerable, the *efficient* setting $\gamma \approx 0.5$ removes 95% of the collisions (leaving $\approx 2\%$) at barely a quarter of the detour cost. The covariance scaling means either choice adapts to the current uncertainty; only the risk appetite remains a design decision.

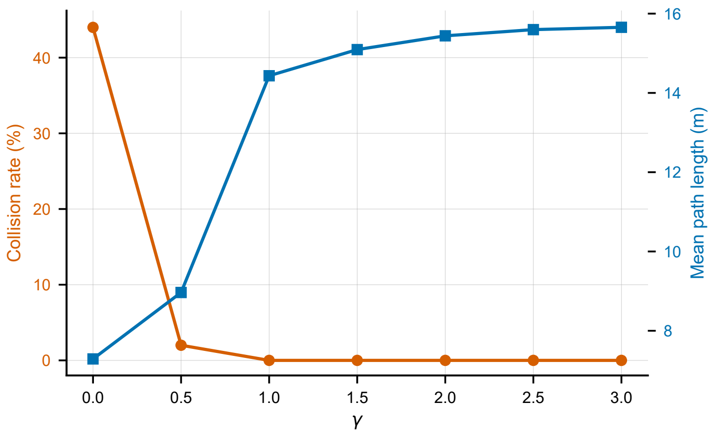

Figure 5. Dynamic obstacle: safety–efficiency trade-off under intermittent visibility. Collision rate (left axis) falls to the oracle floor as γ grows; mean path length (right axis) roughly doubles.

6.5. Ablation: Margin Size vs. Adaptivity

The covariance-aware margin fuses two effects: it is *larger* on average and it is *adaptive* (shaped by $\sqrt{\lambda_{\max}}$). To disentangle which of the two actually drives the safety gain, we run a matched-margin ablation: a *fixed* margin is set to the covariance-aware controller's mean applied inflation, and both controllers are evaluated on identical seeds. Exact matching is not achievable in principle—the margin changes the trajectory, which changes detection and tracking duration, and hence the exposure the other arm realises—so the residual mismatch is measured and reported rather than assumed away.

Table 5 reports the outcome for both scenarios. Against the sensed static obstacle, at matched mean margin (a constant inflation of 0.146 m, the mean the covariance-aware controller applied at $\gamma = 2$), the fixed control recorded the same zero collision count (0 of 50), which is not an equivalence result. Two qualifications apply. The constant was matched to the *all-mission* mean, which (the constraint being active on 93.6% of steps here) is about 7% below the active-step mean, so the fixed arm was mildly under-matched; the comparison is retained as descriptive. And with safety not established equal, the small path difference (6.99 vs. 7.00 m) is not read as an efficiency advantage. What the static arm does establish is that the covariance-aware controller reached a working margin *automatically* from the online covariances, whereas the fixed control required knowing the constant a priori—obtainable here only by first running the covariance-aware controller and measuring it.

The dynamic arm required a correction. A first version of this experiment reported a matched constant of 4.44 m; that figure was an instrumentation artifact (a stale logged value carried over the ≈72% of the mission during which no obstacle constraint existed) and its row is withdrawn. The corrected experiment follows a protocol frozen in the released repository: the matching estimand is the reference arm's mean node-0 inflation *conditional on an active obstacle constraint*, estimated on a calibration seed stream disjoint from the evaluation seeds (frozen at $c = 2.62$ m) and then evaluated on the original paired seeds. The realised active-step means of the two arms agreed to 0.3%, so the arms were genuinely matched to the declared estimand. The outcome is 0/50 collisions for the covariance-aware controller versus 1/50 for the matched constant (exact McNemar $p = 1$): with a single discordant pair, the experiment is *inconclusive* on whether temporal adaptation improves safety or efficiency beyond a calibrated constant, and we report it as such rather than in either direction. The earlier conclusion that margin size alone governs safety is accordingly *suspended*, not reversed. The Wilson 95% intervals also show that a 0/50 outcome still admits an upper bound near 7% at $N_{\text{MC}} = 50$.

Table 5. Matched-margin ablation ($N_{\text{MC}} = 50$). Static: constant set to the all-mission mean inflation (descriptive; ≈7% below the active-step mean). Dynamic: corrected experiment—constant frozen at the reference arm's active-step conditional mean from a disjoint calibration stream; realised exposure matched to 0.3%; the outcome is inconclusive (one discordant pair, McNemar $p = 1$). Collision rate with Wilson 95% interval.

Scenario	Controller	Collision Rate (95% CI)	Path [m]
Static	fixed δ_0	14.0% [7.0, 26.2]	6.93
	fixed-matched (0.146 m)	0.0% [0.0, 7.1]	7.00
	covariance-aware (adaptive)	**0.0% [0.0, 7.1]**	**6.99**
Dynamic	cv_fixed (δ_0)	44.0% [31.2, 57.7]	7.29
	fixed-matched ($c = 2.62$ m, frozen)	2.0% [0.4, 10.5]	16.37
	cv_cov (adaptive)	0.0% [0.0, 7.1]	15.44

6.6. Generalization across Randomized Geometries

The static-obstacle results above, and the sweep that selected $\gamma = 2$, share one fixed layout, which leaves open the possibility that the zero-collision operating point is an artifact of the geometry it was tuned on. We test this directly by freezing $\gamma = 2$ and re-drawing the geometry. For each layout the obstacle is placed within contact distance of a random point on the patrol, so that avoidance is genuinely required, and the three landmark positions are re-drawn under placement constraints (a minimum mutual separation, a minimum standoff from the obstacle, and at least one landmark within sensing range of the start). An oracle feasibility filter then retains only layouts that the ideal-state controller completes collision-free on three probe seeds. This screens out layouts that are infeasible for an ideal-state controller, but it certifies only those three seeds: the oracle still records occasional collisions on other seeds of accepted layouts (Table 6), so acceptance is evidence of solvability rather than a guarantee of it, and a collision in another mode is attributable to estimation or margin rather than to an impossible scenario. Thirty accepted geometries, each run under twenty shared noise realizations for all four modes, give 600 trials per mode.

Table 6 summarizes the outcome and Figure 6 shows sample layouts together with the per-geometry breakdown. Two features of the design deserve emphasis, because both materially affect the numbers. First, the screening and evaluation noise streams are *disjoint*: the three realizations used by the oracle feasibility filter are drawn from a separate stream and never enter the evaluation, so no screening observation appears in the final denominator. Second, collisions are scored *continuously*: the minimum distance from each travelled segment to the obstacle is used rather than the distance at sampled states only, which at up to 0.12 m of travel per step against a 0.30 m keep-out could otherwise miss a grazing pass. Third, trials are *clustered* within geometries, so an interval computed as though the 600 trials were independent would overstate precision: the estimated design effect is 3.7–4.4 for the fixed-margin arms. We therefore report bootstrap intervals over geometries throughout.

The covariance-aware margin collided in 3 of 600 trials (0.50%, cluster-bootstrap 95% CI $[0.00, 1.33]$), confined to 2 of the 30 geometries, with a worst-case penetration of 6.9 cm. It does not eliminate collisions. What it does is reduce them by roughly fifty-fold relative to a fixed margin: the SLAM and odometry fixed-margin controllers collide in 26.8% $[20.3, 33.7]$ and 29.8% $[22.3, 37.3]$ of trials, in 29 and 26 of the 30 geometries respectively. A comparison that does not assume independence among trials *within* a geometry is the geometry-level one: the covariance-aware margin collides less often than the fixed margin in 29 of 30 geometries, never more, and ties in the remaining one (sign test, $p = 3.7 \times 10^{-9}$; this still treats geometries as independent). Because the same 20 noise realizations are reused across all 30 geometries the design is *crossed*, not merely clustered; two-way bootstrap intervals resampling geometries and seeds jointly give pairwise risk differences of -29.3 pp $[-39.0, -20.5]$ against the odometry fixed margin, -26.3 pp $[-35.5, -18.2]$ against the SLAM fixed margin, and $+0.17$ pp $[-1.33, +1.67]$ against the oracle (a failure to detect a difference, not evidence of equivalence).

Because the fixed margins above use the small base buffer δ_0, they leave open whether a *sensibly calibrated* constant would close the gap. We therefore ran a pre-registered follow-up: a single fixed constant was set to the covariance-aware controller's active-step mean inflation, calibrated on ten disjoint training layouts with disjoint seed streams and frozen ($c = 0.273$ m) before any evaluation trial, then evaluated over the identical 30 geometries and 20 evaluation seeds (with a nine-cell bit-exact replay validation against the stored records gating the run). The frozen constant collided in 1 of 600 trials against the adaptive margin's 3 of 600; the paired two-way rate difference is $+0.33$ pp $[-0.67, +1.83]$, spanning zero. Per the interpretation declared in the frozen protocol, the two are *comparable*: the transfer advantage over the δ_0 baselines is carried by margin *size*, and the covariance-aware controller's demonstrated role is that it *generated* that margin automatically—the calibrated constant was obtained by running the adaptive controller on training layouts and summarising it, a procedure that presupposes the adaptive controller. The oracle itself records 0.33%, since it navigates with the small constant buffer $\delta_0 = 0.06$ m and the feasibility filter certifies only three probe seeds, so a tight layout can still produce a graze on an unscreened realization. The operating point tuned on a single geometry therefore transfers, in the sense of a large and consistent reduction rather than an absolute guarantee, to unseen static-obstacle layouts drawn from the same arena and sensing regime, with no per-geometry retuning.

An earlier version of this study reported no collisions for the covariance-aware margin. That result was obtained with screening realizations included in the evaluation set and with clearance scored only at sampled states; correcting both produced the numbers above. We report the correction explicitly because it illustrates how sensitive an apparent zero-event outcome can be to evaluation-protocol details.

One approximation in the corrected scoring should be noted. The continuous test interpolates a straight chord between consecutive sampled poses, whereas the unicycle actually follows a constant-twist arc between them. At these step sizes the chord–arc discrepancy is far smaller than the covariance-aware penetrations reported here (5.9–6.9 cm), so it does not affect that result, but it could matter for millimetre-scale grazes such as the oracle's, whose count should therefore be read as approximate.

Table 6. Randomized-geometry transfer study: static obstacle, γ frozen at 2, 30 oracle-feasible geometries $\times$ 20 evaluation seeds (600 trials per mode), with screening seeds disjoint from evaluation seeds and collisions scored continuously along each travelled segment. Collision rate as a count and percentage with a cluster-bootstrap 95% interval over geometries, worst true clearance, and the number of geometries with at least one collision.

Strategy	Collision Rate (95% CI)	Worst Clearance [m]	Geometries w/ $\geq$1 Coll.
oracle (clairvoyant, fixed δ_0)	2/600, 0.33% [0.00, 1.00]	−0.003	1 / 30
odom + fixed δ_0	179/600, 29.8% [22.3, 37.3]	−0.280	26 / 30
slam + fixed δ_0	161/600, 26.8% [20.3, 33.7]	−0.240	29 / 30
slam + covariance-aware	**3/600, 0.50% [0.00, 1.33]**	−0.069	**2 / 30**

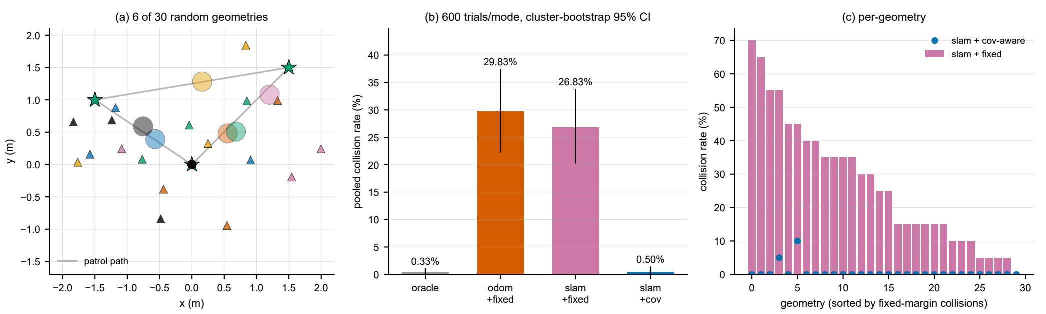

Figure 6. Randomized-geometry transfer study with γ frozen at 2. (**a**) Six of the thirty accepted layouts: patrol path (grey), waypoints (stars), and the obstacle disk and landmark triangles of each layout in a shared colour. (**b**) Pooled collision rates over 600 trials per mode; error bars are the cluster-bootstrap intervals over geometries reported in Table 6. (**c**) Per-geometry collision rates, sorted by the fixed-margin rate: the fixed margin collides in 29 of the 30 geometries, the covariance-aware margin in 2 of the 30.

6.7. Robustness to Monte-Carlo Sample Size

To confirm that the collision results are not artifacts of the trial count, we repeated the principal studies at $N_{\mathrm{MC}} = 100$ (extending the same seed stream, so the first 50 realizations coincide with the primary study). Table 7 reports the collision rates with Wilson 95% confidence intervals. All conclusions are unchanged: the covariance-aware margin recorded no collision against either the sensed static obstacle or the dynamic obstacle (0/100, upper bound 3.7%, matching the oracle's count); the static fixed-matched arm again recorded 0/100 at the same 0.146 m mean margin (descriptive, as in Table 5); and the free-space tracking gain grows slightly to 54% versus odometry (0.062 vs. 0.135 m). Point estimates shift only within the $N_{\mathrm{MC}} = 50$ intervals (for example, dynamic cv_fixed 44% → 42% and static slam-fixed 14% → 15%).

Table 7. Robustness check at $N_{\mathrm{MC}} = 100$: collision rate with Wilson 95% confidence interval. All findings of the $N_{\mathrm{MC}} = 50$ study are reproduced.

Scenario	Strategy	Collision Rate (95% CI)
Static	odom + fixed δ_0	36.0% [27.3, 45.8]
	slam + fixed δ_0	15.0% [9.3, 23.3]
	slam + covariance-aware	**0.0% [0.0, 3.7]**
	fixed-matched (size only)	0.0% [0.0, 3.7]
Dynamic	oracle (clairvoyant)	0.0% [0.0, 3.7]
	static (ignore motion)	89.0% [81.4, 93.7]
	cv_fixed	42.0% [32.8, 51.8]
	cv_cov (covariance-aware)	**0.0% [0.0, 3.7]**

6.8. Robustness to Filter Mis-calibration

The covariance-aware margin is only as trustworthy as the covariance that feeds it, so a natural question is how it behaves when the filter's noise model is wrong. We test this on the sensed static obstacle by scaling the filter's *assumed* process and measurement noise by a factor while drawing the true noise from the original, fixed distribution on the same seeds. The ratio of assumed to true noise variance thus sweeps the filter from overconfident (ratio below one, the reported covariance smaller than the errors warrant) through consistent (ratio one) to conservative (ratio above one); γ is held at 2 and only the filter's self-assessment changes, so this probes genuine (in)consistency rather than a post-hoc rescaling of the reported covariance.

Table 8 and Figure 7 report the outcome. Two features of the results are notable. First, the true localization error is essentially flat across the entire sweep (0.075–0.077 m): the EKF point estimate is largely insensitive to noise mis-tuning, whereas the reported covariance is not—it tracks the true error near the consistent setting and departs from it monotonically, falling below when the filter is overconfident and rising above when conservative, the signature of (in)consistency. Because the margin is sized from that covariance, overconfidence is the failure direction. Second, the margin remains robust across a wide band of it: the covariance-aware controller recorded no collision (0/50) for every ratio down to 0.25—a filter that under-states its variance fourfold—and admits only a single graze (2%, worst penetration 1.6 cm) at a tenfold overconfident filter (ratio 0.10), while the covariance-blind fixed-margin reference collides in 14% on the same obstacle. The worst-case clearance degrades smoothly with overconfidence, from +0.21 m (fourfold conservative) through +0.07 m (consistent) to −0.02 m (tenfold overconfident): as the adaptive term shrinks with the collapsing covariance, the constant base buffer $\delta_0 = 0.06$ m sets a floor, so the mechanism degrades progressively rather than abruptly. Path length varied by under 2% across the sweep. The margin therefore reflects the reliability of its filter: it under-inflates in proportion to the filter's overconfidence, but the fixed floor, together with the mild sensitivity of the point estimate, leaves a broad band of safe operation before that under-inflation degrades safety.

https://doi.org/10.3390/s1010000

Table 8. Robustness to filter mis-calibration ($N_{MC} = 50$, sensed static obstacle, $\gamma = 2$). The filter's assumed noise variance is scaled relative to the true variance; ratios below one make the filter overconfident. Collision rate (Wilson 95% confidence interval), worst true clearance, mean reported pose standard deviation, and true localization error. The covariance-blind fixed-margin reference collides in 14.0% [7.0, 26.2].

Assumed/True Variance	Collision Rate (95% CI)	Worst Clear. [m]	Reported σ [m]	True Loc. Err. [m]
0.10 (10× overconf.)	2.0% [0.4, 10.5]	−0.016	0.022	0.077
0.25 (4× overconf.)	0.0% [0.0, 7.1]	+0.008	0.034	0.076
0.50 (2× overconf.)	0.0% [0.0, 7.1]	+0.037	0.048	0.076
1.00 (consistent)	0.0% [0.0, 7.1]	+0.073	0.067	0.076
2.00 (2× conserv.)	0.0% [0.0, 7.1]	+0.129	0.095	0.075
4.00 (4× conserv.)	0.0% [0.0, 7.1]	+0.206	0.133	0.075

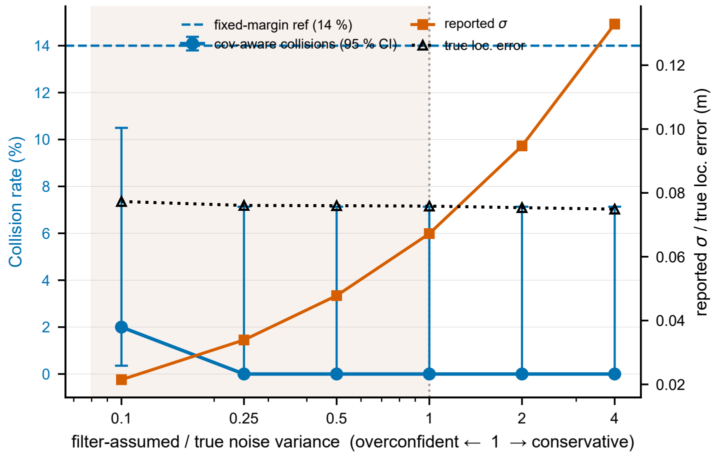

Figure 7. Robustness to filter mis-calibration on the sensed static obstacle, γ held at 2. The horizontal axis is the ratio of the filter's assumed to the true noise variance (left of the dotted line, the shaded region, the filter is overconfident). Left axis: covariance-aware collision rate with Wilson 95% intervals, against the covariance-blind fixed-margin reference (dashed). Right axis: the mean reported pose standard deviation and the true localization error, which stay close near the consistent setting and separate as the filter is mistuned.

6.9. Computational Performance

Per-step computation was measured against the controller sampling period on the unknown-map controllers. All timings were collected on a 13th-generation Intel Core i9-13900K (Windows 11) under MATLAB R2022a with CasADi v3.5.5 and IPOPT at default solver settings, measured as per-step wall-clock over all Monte-Carlo trials of the timing runs; no solver failure or fallback activation was recorded in these runs. In free space the NMPC solves in 6.1 ms on average (95th percentile 8.3 ms) against a 50 ms period, an ≈12% duty cycle. The parameterised-obstacle controller, a larger multiple-shooting problem with nonconvex constraints and slack variables, averages 11.5 ms (95th percentile 14.9 ms) against a 200 ms period, an ≈6% duty cycle. The EKF-SLAM and obstacle-estimator updates cost well under a millisecond, and the covariance term in (7) is a single small eigenvalue problem; all are negligible. The framework therefore runs with roughly an order-of-magnitude real-time margin, and the covariance-aware safety mechanism adds essentially no computational cost.

7. Discussion

The experiments together support a qualified conclusion. Under certainty equivalence, feeding SLAM into the loop sharpens accuracy and softens the severity of collisions, yet it did not, by itself, eliminate the observed collisions; the zero-collision counts in these trials were obtained only with an enlarged margin, whether adaptive or constant. The corrected matched-margin ablation (Table 5) is inconclusive on whether temporal adaptation adds safety beyond a calibrated constant of the same active-step size, and the earlier conclusion that margin size alone governs safety remains suspended. The merit of sizing the margin from the live covariances is therefore not a categorical gain in safety but two practical ones. First, it self-tunes the keep-out online, sparing the designer from hand-picking a safe constant a priori—a constant that, in our experiments, was obtainable only by first running the covariance-aware controller and measuring what it did. Second, the margin follows the live covariance wherever the mission takes it, requiring no knowledge of where the risky stretch lies; in this workspace the inflation stayed small everywhere, so against the sensed static obstacle the observed zero-collision count cost less than 1% of path. Dead reckoning can do neither, because it lacks any reliable estimate of its own uncertainty.

The unknown-map setting also revealed three lessons that a fully known world hides. First, an uncertainty-aware margin is only as *well-calibrated* as the filter that feeds it. EKF-SLAM is known to become overconfident—its covariance shrinking faster than the true error—as a consequence of linearizing about an uncertain estimate under weak observability [9,10]; we observed precisely this in stretches where a single landmark was in range—the heading is then weakly observable, and an unlucky initialization can leave the estimate biased well beyond what the covariance admits. When the filter is overconfident, the margin under-inflates in proportion. The mis-calibration sweep of Section 6.8 quantifies how much this costs: no collision is observed until the filter under-states its variance roughly fourfold, and even at tenfold overconfidence the constant base buffer holds the failure to a single shallow graze—so the effect is real but bounded. Second, as the dynamic study showed, the margin is only as reliable as the lifetime of the estimate feeding it: without track management, a coasting track's unbounded covariance inflates the margin until the robot can no longer move. Third, safety and perception are *coupled through the trajectory*: the evasive detours that improve safety also take the robot away from its landmarks, degrading the very localization that mission precision depends on, so waypoints are attained in the belief frame while the true pass drifts wide (Table 4). The nominal margin grows with the same covariance the detour inflates, and no collision was recorded in these trials, but the observation argues for perception-aware detouring, i.e., resolving avoidance ambiguity toward landmark coverage, as a natural refinement.

Several modeling choices bound the scope. Landmarks are assumed distinguishable (known data association), which suits uniquely identifiable features or beacons but sidesteps the association failures of cluttered environments. The margin (7) uses a Gaussian, single-obstacle tightening and assumes a reasonably calibrated filter; it is chance-constraint-inspired rather than a calibrated probability bound, and a formal bound would require accounting for linearization and correlation between obstacles. The scope of what is and is not established should therefore be stated without ambiguity. The method carries no calibrated collision-probability bound, no recursive-feasibility proof, no stability guarantee, and no certified fail-safe: the obstacle constraint is soft, and the evidence reported here is a finite set of Monte-Carlo outcomes. Every safety statement in this paper is a statement about observed trials with its associated interval, not about a guaranteed property, and the contribution is accordingly the online *scheduling* of a safety margin from live covariance; the corrected matched-margin ablation (Table 5) is inconclusive on whether temporal adaptation adds safety beyond a calibrated constant of the same active-step size, and no such claim is made in either direction. The study is in simulation with three landmarks and a single obstacle per scenario. The randomized-geometry study (Section 6.6) shows that the frozen γ transfers across static-obstacle layouts drawn from the same arena and sensing regime; transfer beyond that family—other arenas, obstacle sizes, or the dynamic scenario—remains untested. The controller uses certainty equivalence for the tracking term (only the obstacle constraint is made uncertainty-aware). The sparse map is also a deliberately demanding case: landmark density governs how quickly the pose covariance grows between re-detections, so a denser map would blunt the very nonstationarity the margin responds to, while a sparser one would sharpen both the drift and the perception–control coupling reported below; a systematic density sweep is left to future work.

Natural extensions include multiple simultaneous obstacles, tube-MPC bounds for the tracking objective, consistency-preserving estimators as the margin's information source, hardware validation on a differential-drive or car-like platform, and a formal chance-constraint derivation that propagates the predicted covariance along the horizon.

8. Conclusions

This study has presented a closed-loop framework that couples EKF-SLAM with a constrained nonlinear MPC for a nonholonomic mobile robot navigating an environment that is unknown before it is sensed—landmarks initialized from first detection, obstacles discovered and estimated online—and, going beyond certainty equivalence, feeds the live covariances back into the controller to size the obstacle keep-out constraints. On a multi-leg patrol in Monte-Carlo simulation, SLAM-in-the-loop reduced terminal error by 48% relative to odometry while building the map from scratch to 4 cm accuracy, and the proposed covariance-aware safety margin recorded no collision with a sensed static obstacle (0 of 50 trials) for less than 1% additional path; the nominal margin is state-dependent, entering a softened constraint that binds only near obstacles. A corrected matched-margin ablation, with the constant frozen from a disjoint calibration stream and realised exposure matched to 0.3%, was inconclusive on whether temporal adaptation adds safety beyond a calibrated constant (0/50 vs. 1/50, one discordant pair); the demonstrated value of the covariance-aware formulation is that it generates working margins online without a hand-selected constant. A randomized-geometry study confirmed that this behaviour generalizes, though not absolutely: with the scaling factor frozen, and with screening seeds held disjoint from evaluation seeds and collisions scored continuously, the margin collided in 0.50% of trials across thirty unseen, oracle-feasible layouts; a fixed constant calibrated from the adaptive controller on disjoint training layouts performs comparably (1 of 600), so the transfer advantage over the small-buffer baselines reflects margin size, generated automatically by the covariance-aware controller (3 of 600, in 2 of 30 geometries) against 27–30% for fixed margins, a roughly fifty-fold reduction rather than elimination. It also tolerated filter mis-calibration over a wide band, recording no collision until the filter understated its own variance roughly fourfold and degrading gracefully beyond that. A sweep of the scaling factor characterized the safety–efficiency trade-off and justified the operating point, and timing measurements confirmed real-time feasibility with an order-of-magnitude margin. The same covariance-aware mechanism extends to a moving, intermittently visible obstacle tracked by a coasting constant-velocity filter, where it recorded the same observed collision count as the clairvoyant oracle (0 of 50 trials vs. 44–88% for fixed-margin strategies) at the price of a more cautious path and of some true-frame waypoint precision, since the evasive detours themselves degrade localization, and provided stale tracks are dropped, without which unbounded prediction uncertainty sacrifices mission completion for safety. The results indicate that the uncertainty already computed by the filters is a valuable, underused signal for safe predictive control in unknown environments.

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Data Availability Statement: The MATLAB/CasADi implementation, the randomized-geometry generator, the accepted layouts and per-geometry outcomes of the randomized-geometry and filter-miscalibration studies (`um_randgeom_v2_results.mat`, `um_consistency_results.mat`; the superseded first-protocol dataset `um_randgeom_results.mat` is retained for provenance and should not be used), the aggregate result summaries of every other study, the clustered-analysis script that produces the intervals reported in Table 6 and Figure 6, the figure-generation scripts, and the compiled manuscript are openly available at https://github.com/11leoool/Uncertainty-Aware-Closed-Loop-Model-Predictive-Control-with-EKF-SLAM (MIT license). Random seeds are fixed and stated in each runnable script, so the per-trial outcomes underlying the aggregate summaries regenerate deterministically from the released code.

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